Mathematical notes on executable common ascent, cross-view invariants, and decision-space uncertainty

RefCloth exploration notes

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Scope. These notes specify three falsifiable source-side candidates. Their ingredients—linear-programming duality, multiobjective ascent, factorial/Möbius contrasts, and ellipsoidal robust optimization—are standard mathematics, not claimed new general theorems. The application hypotheses must be distinguished from these identities. No experimental outcomes are reported here; no result below establishes task failure, closed-loop transfer, or an unconditional deployment certificate.

1 Common ascent in the final executable box

Fix a texture-parameter vector $\theta \in [-b, b]^p$, step budget $\eta > 0$, and source task–state objectives J_i, $i = 1, \dots, m$. Put $g_i = \nabla J_i(\theta)$ and $q_i = g_i / s_i$, where $s_i > 0$ is fixed for this step. Raw gradients use $s_i = 1$; another choice is $s_i = \max(\|g_i\|_1, \varepsilon)$. Zero-gradient groups must be retained. Common regularizers belong in every J_i if their gradients enter the update.

An update followed by parameter clipping can execute precisely the increments

$$B = \{\delta : \ell_j \le \delta_j \le u_j\}, \quad \ell_j = \max\{-\eta, -b - \theta_j\}, \qquad u_j = \min\{\eta, b - \theta_j\}. \tag{1}$$

This box contains zero but need not be symmetric. With fixed offsets c_i, solve

$$\gamma^* = \max_{\delta \in B} \min_i (q_i^\top \delta + c_i) = \max_{\delta, t} \{t : q_i^\top \delta + c_i \ge t \ \forall i, \ \delta \in B\}. \tag{2}$$

The executable output is the *primal increment* δ^* itself. Normalization and execution geometry are distinct design choices.

Proposition 1.1 (Support-function dual)**.** *For* $\Delta_m = \{w \ge 0 : \sum_i w_i = 1\}$ *and* $h(w) = \sum_i w_i q_i$,

$$\gamma^* = \min_{w \in \Delta_m} \left\{ \sum_{j=1}^{p} \max\{\ell_j h_j(w), u_j h_j(w)\} + w^\top c \right\}. \tag{3}$$

Every feasible δ *gives a lower bound in* (2)*; every simplex weight vector gives an upper bound in* (3)*. Their difference is a directly checkable primal–dual gap.*

Proof. Replace the finite minimum over groups by the minimum over simplex mixtures. The resulting objective is affine in each variable on convex compact sets, so minimax interchange applies. Maximizing $h^\top \delta$ over B separates coordinate-wise, giving the support function displayed in (3). □

For the full symmetric cube $B = [-\eta, \eta]^p$ and $c = 0$,

$$\gamma^* = \eta \min_{w \in \Delta_m} \left\| \textstyle\sum_i w_i q_i \right\|_1. \tag{4}$$

Thus $\gamma^* = 0$ exactly when $0 \in \operatorname{conv}\{q_i\}$ in this symmetric case. More generally, with $c = 0$, $\gamma^* > 0$ is equivalent to existence of a strict common-ascent increment *in the actual box*. The convex-hull equivalence cannot be transferred without qualification to a clipped box with zero on its boundary.

Multiobjective gradient combination predates this application [1, 2]. What must be tested is whether respecting the executed box changes native updates, beyond using existing gradient combination alone.

Why a final sign can invalidate a common-ascent direction

Take two linear objectives with $g_1 = (1, 0)$, $g_2 = (-2, 1)$ and $B = [-1, 1]^2$. The sign of their mean is $(-1, 1)$, with objective gains $(-1, 3)$. Projecting each conflicting gradient against the original other gradient, as in the two-group PCGrad calculation [2], gives

$$\widetilde{g}_1 = (1/5, 2/5), \qquad \widetilde{g}_2 = (0, 1).$$

Their sum has positive gains $(1/5, 1)$, whereas its sign $(1, 1)$ has gains $(1, -1)$. A strict common-ascent direction existed and was destroyed by signing. The maximin primal solution instead is

$$\delta^* = (1/3, 1), \qquad \gamma^* = 1/3.$$

Indeed, gains at least t require $\delta_1 \geq t$ and $\delta_2 \geq t + 2\delta_1 \geq 3t$, hence $t \leq 1/3$. Even signing the optimal dual combination is insufficient: weights $(2/3, 1/3)$ give $h = (0, 1/3)$, whose sign $(0, 1)$ has worst gain zero. The missing fractional coordinate is determined by active primal constraints. For ℓ_1-normalized rows $(1, 0)$ and $(-2/3, 1/3)$, the sign of their mean still harms the second group; the normalized LP optimum is $(1/5, 1)$.

Proposition 1.2 (Sparse fractionality of an optimal basic solution)**.** *After eliminating fixed coordinates of B, there exists an optimal basic solution with at most $m - 1$ increment coordinates strictly inside their bounds. If that solution has k active group constraints, the sharper limit is $k - 1$.*

Proof. There are $p + 1$ free LP variables (δ, t). If r increment coordinates are interior, at most $p - r$ independent box constraints are active. A basic solution requires $p + 1$ independent active constraints, so $p - r + k \geq p + 1$. An optimal vertex exists for this pointed LP with bounded increment box and finite optimum. The claim concerns an optimal basic solution, not every point on its optimal face or an averaged solver iterate. □

Proposition 1.3 (Conditional native-export gain bound)**.** *Let $F_i(\theta) = \mathcal{F}_i(T_\theta)$ be a continuously relaxed native objective. At the current point and along the relevant box segment, assume*

$$\|\nabla F_i(\theta) - g_i\|_1 \leq \rho_i, \qquad \|\nabla F_i(a) - \nabla F_i(a')\|_1 \leq L_i \|a - a'\|_\infty .$$

Suppose $\mathcal{F}_i$ is κ_i-Lipschitz in texture ℓ_∞ norm, and the actual export Q satisfies $\|Q(T) - T\|_\infty \leq q$. Then

$$\mathcal{F}_i(Q(T_{\theta+\delta})) - \mathcal{F}_i(Q(T_\theta)) \geq g_i^\top \delta - \rho_i \eta - \tfrac{1}{2} L_i \eta^2 - 2\kappa_i q. \tag{5}$$

Proof. Integrate the native gradient along the step. Hölder's inequality bounds its initial discrepancy by $\rho_i \eta$, and the integrated remainder by $L_i \eta^2/2$. Comparing quantized and relaxed textures at both endpoints costs at most $2\kappa_i q$. □

Dividing F_i, g_i and the remainder by the same fixed s_i, the LP offset $c_i = -(\rho_i \eta + L_i \eta^2/2 + 2\kappa_i q)/s_i$ gives a conditional common normalized native-gain lower bound γ^*.

These bounds require justified gradient-error, curvature, and export constants. Finite probes do not establish uniform constants. Nearest 8-bit rounding has $q \leq 1/(2 \cdot 255)$; truncation may require $q \leq 1/255$. Identical exported PNGs have zero native gain despite a positive relaxed prediction. Clipping and nonsmooth TV also require their own smoothness qualifications.

2 Four-corner interaction as a conditional rendering invariant

Fix simulator state, instruction, proprioception, inference noise, and action coordinates. Let $A_P(u, v) \in \mathbb{R}^d$ denote the selected action vector when the two views use textures u, v through path $P \in \{R, N\}$ (synthesis/native). Physical deployment uses the shared diagonal $u = v = \theta$; off-diagonal evaluations are camera-input interventions, not separately deployable textures. Put $A_{P,ij} = A_P(i\theta, j\theta)$ and

$$\begin{aligned} d_{1,P} &= A_{P,10} - A_{P,00}, & d_{2,P} &= A_{P,01} - A_{P,00}, \\ d_{J,P} &= A_{P,11} - A_{P,00}, & I_P &= A_{P,11} - A_{P,10} - A_{P,01} + A_{P,00}. \end{aligned} \tag{6}$$

All four corners within each contrast use the same rendering path.

Proposition 2.1 (Unique normalized linear invariant)**.** *A linear contrast* $\sum_{i,j\in\{0,1\}} c_{ij} A_{ij}$ *with* $c_{11} = 1$ *is invariant to every additive nuisance* $b + e_1(i) + e_2(j)$ *if and only if*

$$(c_{00}, c_{10}, c_{01}, c_{11}) = (1, -1, -1, 1). \tag{7}$$

Proof. Annihilating the first-view indicator forces $c_{10} + c_{11} = 0$; annihilating the second forces $c_{01} + c_{11} = 0$; annihilating constants forces $\sum c_{ij} = 0$. These equations give (7), which also suffices. The additive nuisance subspace has dimension three in the four-dimensional corner space, leaving one quotient coordinate. This is a standard factorial/Möbius contrast, not a new general invariance theorem [3]. □

Define the cross-path action error $E(u, v) = A_R(u, v) - A_N(u, v)$. Exactly,

$$I_R - I_N = E(\theta, \theta) - E(\theta, 0) - E(0, \theta) + E(0, 0). \tag{8}$$

Thus the error cancels if $E(u, v) = b + e_1(u) + e_2(v)$ at these corners, even if $d_{J,R} - d_{J,N}$ is large. A nonlinear fusion error need not be separable. Replacing only $A_{R,00}$ by $A_{N,00}$ changes the contrast to $I_R - E(0, 0)$ and spoils the cancellation of a constant cross-path offset.

Proposition 2.2 (Behavioral relevance requires unilateral control)**.** *For any norm,*

$$\|d_{J,N}\| \geq \left[\|I_N\| - \|d_{1,N}\| - \|d_{2,N}\|\right]_+ . \tag{9}$$

If $\|I_R - I_N\| \leq \xi$ *and* $\|d_{v,N}\| \leq \tau_v$*, then* $\|d_{J,N}\| \geq [\|I_R\| - \xi - \tau_1 - \tau_2]_+$.

Proof. Use $d_{J,N} = I_N + d_{1,N} + d_{2,N}$ and the reverse triangle inequality twice. □

RMS on a fixed d-coordinate subset is a scaled Euclidean norm, so the same bound applies. Constraints on synthesis unilateral effects alone do not give native unilateral control without an additional cross-path error bound.

Identification limit. Even knowing the native clean corner does not identify native total joint displacement under unrestricted additive error. Let the known native clean output be $\bar{a}_{N,00}$ and set $b = A_R(0, 0) - \bar{a}_{N,00}$. For any action vector z, the family $A_N^z(i, j) = A_R(i, j) - b - iz$ matches that same known clean output and preserves the interaction while changing native joint displacement by $-z$. Consequently an invariant interaction is not a surrogate for the entire native action effect without additional calibration or structural assumptions.

Mixed curvature, relative error, and protected local updates

For an ideal C^2 extension on the rectangle of texture interpolations, two applications of the fundamental theorem of calculus give

$$I_P(\theta) = \int_0^1 \int_0^1 D_{12} A_P(s\theta, t\theta)[\theta, \theta], dt\, ds. \tag{10}$$

If the bilinear operator norm of $D_{12}E$, for the chosen input and output norms, is uniformly at most L_{12} there,

$$\|I_R - I_N\| \le L_{12} \|\theta\|^2 . \tag{11}$$

This is an *absolute* second-order error bound. The interaction itself is generally second-order, so relative fidelity need not improve as the texture shrinks. For example, scalar maps $A_N(u, v) = uv$ and $E(u, v) = auv$ give $I_N(\theta) = \theta^2$ and $|I_R - I_N|/|I_N| = |a|$ for every nonzero θ. When $I_N = 0$, relative error is undefined. A fixed denominator floor must be specified before inspection if relative errors are summarized.

Smoothness limit. Quantized PNGs, finite-precision inference, clipping, and visibility changes do not automatically admit this C^2 extension. The four-corner algebra, uniqueness, and triangle inequalities remain exact for observed outputs; the smooth mixed-curvature interpretation is conditional. Uncoupled inference noises would additionally confound the four-corner contrast.

A possible protected source objective is

$$\max_{\theta} \ \mathbb{E} \|I_R(\theta)\|^2 , \qquad C_v(\theta) = \mathbb{E} \|d_{v,R}(\theta)\|^2 \le \tau_v^2, \qquad T_\theta \in \mathcal{T}_\epsilon. \tag{12}$$

An average constraint does not imply each-state protection; a worst-state or tail claim requires the corresponding prespecified constraint. Native unilateral effects must be tested separately, as (9) requires. For $g = \nabla \|I_R\|^2$ and $h_v = \nabla C_v$, an economical local problem is

$$\max_{\delta \in \mathcal{C},\ \|\delta\|_2 \le r} g^\top \delta, \qquad \mathcal{C} = \{\delta : h_v^\top \delta \le 0,\ v = 1, 2\}. \tag{13}$$

Its solution is $r\Pi_{\mathcal{C}}(g)/\|\Pi_{\mathcal{C}}(g)\|$ when the projection is nonzero, and zero otherwise. Moreau's cone decomposition gives the corresponding support-function identity. This is standard constrained gradient projection, related to GEM [4]; the two-constraint dual can be solved by enumerating active sets. If ∇C_v is β_v-Lipschitz, $C_v(\theta + \delta) \le C_v(\theta) + \beta_v r^2/2$ for such a first-order feasible step. Finite-step protection still needs a margin or an actual backtracking check.

Local spectral observation. For a fixed action direction a, let $H_a = \mathrm{sym}(D_{12}(a^\top A)(0, 0))$. With bounded third derivatives,

$$a^\top I(tv) = t^2 v^\top H_a v + O(t^3).$$

If P projects onto $\ker D_1 A(0, 0) \cap \ker D_2 A(0, 0)$, maximizing this signed quadratic over unit vectors in that subspace is a restricted eigenvalue problem. Maximizing the full vector norm is instead quartic and is not solved by one such eigenvector. First-order unilateral nulling does not remove second-order solo changes. Also, $\|I(\theta)\|^2 = O(\|\theta\|^4)$ near zero, so its gradient can vanish there; initialization and optimization gains must be separated.

3 Decision-space ellipsoidal renderer uncertainty

Fix a source action head H and a linear map P selecting execution-relevant coordinates in one common control-unit system. Let $z_{N,0}$ be native clean features, $d_R = z_R(\theta) - z_R(0)$ and $d_N = z_N(\theta) - z_N(0)$. Define

$$u_R = P\{H(z_{N,0} + d_R) - H(z_{N,0})\}, \quad u_N = P\{H(z_{N,0} + d_N) - H(z_{N,0})\}, \quad e = u_N - u_R. \tag{14}$$

The action head is evaluated as given, without a linear approximation. The native clean anchor is fixed. Production affine action-unnormalization scales belong in P; their offsets cancel in displacements. Mixing normalized and unnormalized coordinates would change the Euclidean objective and its geometry.

Given a fitted $C \succ 0$, consider the conservative action-energy objective

$$V(u) = \min_e \{\|u + e\|_2^2 : e^\top C^{-1} e \le r(u)^2\}. \tag{15}$$

Absolute uncertainty uses $r(u) = \rho$. Relative uncertainty uses $r(u) = \rho \|u\|_2$, with a separately estimated dimensionless shape. The following is a standard ellipsoidal quadratic subproblem [5, 6]; its exact solution is useful for specifying the gradient.

Proposition 3.1 (Inner optimum and envelope derivative)**.** *Write $C = U \operatorname{diag}(c_j) U^\top$, $v = U^\top u$, and suppose $r(u) > 0$. If $u^\top C^{-1} u \le r(u)^2$, then $V(u) = 0$ and $e^* = -u$. Otherwise there is a unique $\mu > 0$ satisfying*

$$\sum_j \frac{c_j v_j^2}{(c_j + \mu)^2} = r(u)^2, \tag{16}$$

and the worst-case remaining action and objective are

$$y^* = u + e^* = U \operatorname{diag}\left(\frac{\mu}{c_j + \mu}\right) v, \qquad V(u) = \|y^*\|_2^2 . \tag{17}$$

At an exterior point, the two envelope gradients are

$$\nabla V_{\text{abs}}(u) = 2y^*, \qquad \nabla V_{\text{rel}}(u) = 2\{y^* - \mu\rho^2 u\}. \tag{18}$$

Proof. The convex inner problem has a unique optimizer. Its Lagrangian is $\|u + e\|^2 + \mu(e^\top C^{-1} e - r(u)^2)$. Exterior points satisfy Slater's condition and have an active constraint. Stationarity gives $e^* = -C(C + \mu I)^{-1} u$, yielding (16)–(17). The left side of (16) decreases continuously from a value above $r(u)^2$ to zero, hence has a unique positive root. Differentiating the Lagrangian with respect to u at its optimum gives $2y^* - \mu \nabla(r(u)^2)$, which proves (18). □

At strict interior points, both value and gradient are zero. With zero radius, the limiting problem is simply $V(u) = \|u\|^2$. Relative uncertainty is positively homogeneous: $V_{\text{rel}}(au) = a^2 V_{\text{rel}}(u)$ for $a \ge 0$. Omitting the second term in its envelope gradient gives the right forward score but the wrong update. A single isotropic subtraction $(\|u\| - \rho)_+^2$ is only a monotone radial transformation of action energy; on one sample its nonzero sign-gradient direction is unchanged. Anisotropic uncertainty must be distinguished from such a scalar reweighting.

What calibration and a conditional lower bound do not prove

If the *actual* error $u_N - u_R$ belongs to the chosen uncertainty set,

$$\|u_N\|^2 \geq V(u_R), \qquad 0 \leq V(u_R) \leq \|u_R\|^2 . \tag{19}$$

This follows because the actual error and zero error are feasible competitors for the inner minimum. If the new texture obeys that membership condition and its lower bound exceeds a measured old native action energy, an improvement of that native action energy follows. Neither an increase of the lower bound alone nor membership measured at a different texture establishes the premise.

A source-only estimation recipe can use disjoint *state groups* for shape and radius estimation. For absolute errors, one possible origin-centered shape is $C = \mathbb{E}[ee^\top] + \lambda I$; centering would remove a systematic error that the uncertainty model may need to cover. Relative errors can instead fit the shape of $e/\|u_R\|$, with a declared treatment of zero or negligible denominators. A held-out empirical quantile of Mahalanobis lengths defines an empirical radius, not a uniform error bound. Correlated probes and state shift further limit a naive probabilistic interpretation. Optimization-selected textures need not follow the calibration distribution, so random-probe coverage cannot certify them.

For a fixed comparison, signal-versus-error covariance can also define generalized directions $Sa = \lambda Ca$, with $S = \mathbb{E}[u_N u_N^\top]$. That is an established generalized-eigenvalue construction. Direct action objectives are already used in VLA texture attacks [7]. Thus a meaningful distinction must be a tested advantage of the uncertainty geometry over both direct action energy and a fixed action-space quadratic metric.

Evaluation requirements implied by the mathematics. An ellipsoid can contain $-u_R$, making the robust score and its gradient zero; this is a substantive failure of optimization usability. Flat-score states must not silently disappear from rank summaries, nor may arbitrary argmax ties be interpreted as useful selection. Report tie rules, degeneracy, and each-state behavior. All candidate updates must be evaluated in the same unweighted native control coordinates. Differing loss scales must not silently change regularization strength. Finally, native value fidelity and native gradient fidelity are different questions: a smaller prediction error does not by itself justify a better executable update.

References

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